

PROJECT DOCUMENTATION

IMPORT DATA USING TRANSFORM MAPS (SPREADSHEET)

Details	Information
Project Title	Import Data Using Transform Maps (Spreadsheet) in ServiceNow
Team ID	SWTID-2026-1408
Team Size	4
Team Leader	SWETHA R
Team Member	AJITHA A
Team Member	AFREEN BENASIR M
Team Member	RIHANA M

1. Introduction

1.1 Project Description

This micro project focuses on importing structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario in which bulk employee data is received in Excel format and migrated into ServiceNow efficiently and accurately.

The spreadsheet data is first loaded into an Import Set table, which acts as a staging area. A Transform Map is then configured to map source fields to the appropriate fields in the Employee Test table. Coalesce is used to identify existing employee records and prevent duplicate records during repeated imports. New employee records are inserted, while existing records can be updated when their information changes. Reports and a dashboard are created to provide meaningful insights into the employee data.

1.2 Project Objectives

1. Understand the ServiceNow data import process.
2. Import employee data from a spreadsheet.
3. Create a custom Employee Test table.
4. Create and use an Import Set table.
5. Configure and use a Transform Map.
6. Map source fields to target fields.
7. Use Coalesce to prevent duplicate employee records.
8. Validate imported and transformed records.
9. Create reports and a dashboard for employee data.

2. Project Overview

2.1 Purpose

The purpose of this project is to provide an efficient method for importing and managing bulk employee information in ServiceNow. Instead of entering employee records manually, the data can be prepared in a spreadsheet and imported using Import Sets and Transform Maps.

Overall Process:

Excel Spreadsheet → Load Data → Import Set Table → Transform Map → Field Mapping → Coalesce → Employee Test Table → Reports → Dashboard

2.2 Key Features

- Import employee data from Excel.
- Create a custom Employee Test table.
- Create an Import Set table.
- Configure a Transform Map.
- Automatically map matching fields.
- Manually map fields using Mapping Assist.
- Transform imported records.
- Validate imported records.
- Prevent duplicates using Coalesce.
- Create reports and a dashboard.

2.3 Project Workflow

Prepare Spreadsheet → Create Employee Test Table → Load Spreadsheet Data → Create Import Set → Create Transform Map → Map Source and Target Fields → Transform Data → Validate Records → Enable Coalesce → Import Updated/New Data → Verify Insert/Update/Ignore → Create Reports → Create Dashboard

3. Architecture

This project is developed completely within ServiceNow and does not require a traditional React frontend, Node.js backend, or MongoDB database architecture. It follows the ServiceNow Import Set and Transform Map architecture.

3.1 System Architecture

External Data Source: Excel Spreadsheet



Load Data



Import Set Table: u_employee_import



Transform Map: Sample Spreadsheet Import



Field Mapping + Coalesce



Employee Test Table: u_employee_test



Reports + Dashboard

3.2 Spreadsheet

The spreadsheet is the external source of employee data. The sample file is named Sample Spreadsheet.xlsx.

Employee ID	Employee Name	Email	Department	Location
-------------	---------------	-------	------------	----------

sbc001	user1	user1@gmail.com	servicenow	tamilnadu
sbc002	user2	user2@gmail.com	servicenow	tamilnadu
sbc003	user3	user3@gmail.com	servicenow	tamilnadu
sbc004	user4	user4@gmail.com	servicenow	tamilnadu
sbc005	user5	user5@gmail.com	salesforce	tamilnadu

3.3 Import Set

The Import Set provides a temporary staging area for spreadsheet data. The imported data is stored in the Employee Import table with backend name u_employee_import.

3.4 Transform Map

The Transform Map controls the movement of data from the Import Set table to the target table.

Property	Value
Transform Map	Sample Spreadsheet Import
Source	Employee Import
Target	Employee Test

3.5 Target Table

The final employee information is stored in the Employee Test table with backend name u_employee_test.

- Employee ID
- Employee Name
- Email
- Department
- Location

3.6 Coalesce

Employee ID is used as the identifying field. If the Employee ID exists, the existing record is updated; if it does not exist, a new record is inserted.

4. Setup Instructions

4.1 Prerequisites

- ServiceNow instance.
- Valid ServiceNow credentials.
- Appropriate ServiceNow permissions.
- Google Sheets or Microsoft Excel.
- Employee sample data.
- Basic knowledge of ServiceNow tables, Import Sets, and Transform Maps.

4.2 Phase 1 – Prepare the Spreadsheet

10. Open Google Sheets or Microsoft Excel.
11. Create a new spreadsheet.
12. Enter the employee data with the required headings: Employee ID, Employee Name, Email, Department and Location.
13. Download the spreadsheet as Sample Spreadsheet.xlsx.

5. Create Custom Employee Table

5.1 Navigation

All → Tables → Create New

5.2 Table Details

Property	Value
Label	Employee Test
Name	u_employee_test

Save the table.

5.3 Create Employee Fields

After saving the table, open the form and use Configure → Form Layout. Create the following fields:

Column Label	Type
Employee ID	String
Employee Name	String
Email	String
Department	String
Location	String

Save and verify that all fields appear on the Employee Test form.

6. Create Import Set Table

6.1 Navigation

All → System Import Sets → Load Data

6.2 Import Details

Upload the prepared Excel spreadsheet. Create or select the Import Set table.

Property	Value
Label	Employee Import
Name	u_employee_import

Click Submit. After the import is completed, select Create Transform Map.

7. Create Transform Map

7.1 Transform Map Details

Property	Value
Name	Sample Spreadsheet Import
Source Table	Employee Import
Target Table	Employee Test

7.2 Auto Map Matching Fields

Click Auto Map Matching Fields. ServiceNow automatically maps source and target fields with matching names.

7.3 Mapping Assist

Use Mapping Assist to verify or manually create the following field mappings:

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

7.4 Transform the Data

14. Select the required Import Set and Transform Map.
15. Click Transform.
16. Allow ServiceNow to process the imported records.
17. Verify the successful transformation message.
18. Open the Employee Test table and verify the records.

8. Running the Application / Project

Unlike a React or Node.js application, this ServiceNow project does not require commands such as `npm start`. The project runs directly inside the ServiceNow instance.

Execution Process:

Login to ServiceNow → Open Load Data → Upload Spreadsheet → Create/Select Import Set → Create Transform Map → Map Fields → Run Transform → Verify Employee Test Table

9. Data Transformation and Validation

After transformation, open the Employee Test table through the Application Navigator. The imported employee records should be displayed.

Expected Result

All valid spreadsheet records should be successfully created in the Employee Test table, with the corresponding Employee ID, Employee Name, Email, Department and Location values.

10. Authentication and Authorization

The project uses ServiceNow's built-in authentication and authorization mechanism. Users must log in with valid credentials. Access to tables, Import Sets, Transform Maps, Reports and Dashboards depends on assigned roles and permissions.

- Upload data.
- Create Import Sets.
- Create Transform Maps.
- Modify employee records.
- Create reports.
- View dashboards.
- Share dashboards.

No custom JWT or token-based authentication is required because the project operates within the ServiceNow platform.

11. Coalesce and Duplicate Prevention

11.1 Purpose

When the same employee spreadsheet is imported multiple times, duplicate records can be created if there is no duplicate-handling mechanism. Coalesce solves this problem by using a unique identifying field.

11.2 Configuration

Open Sample Spreadsheet Import → Field Maps → Employee ID → Set Coalesce = True → Save the Transform Map.

11.3 Coalesce Logic

Employee ID exists? → Yes: Update existing record. → No: Insert new record.

This allows the system to distinguish between new and existing employees during repeated imports.

12. Testing Insert, Update and Ignore Operations

12.1 Load the New Spreadsheet

19. Navigate to All → System Import Sets → Load Data.
20. Select Existing Table → Employee Import.
21. Choose the new Excel file.
22. Set Sheet Number = 1 and Header Row = 1.
23. Click Submit.

12.2 Run Transform

Select Run Transform, choose the existing Transform Map and click Transform. After completion, open Transform History.

12.3 Expected Result

Operation	Records
Inserted	2
Updated	2
Ignored	0
Total	4

The two new Employee IDs are inserted and the two existing Employee IDs are updated.

12.4 Re-import the Same File

Upload the exact same spreadsheet again using the same Import Set and Transform Map.

Operation	Records
Inserted	0
Updated	0
Ignored	4
Total	4

This demonstrates duplicate prevention for repeated imports.

13. User Interface

- Spreadsheet – Contains the employee data to be imported.
- Employee Test Table – Displays final employee records.
- Load Data Page – Used to upload the Excel spreadsheet.
- Transform Map – Configures source, target and transformation.
- Field Mapping – Displays source-to-target field relationships.
- Transform History – Displays inserted, updated, ignored and failed records.
- Reports – Provides graphical and list-based views.
- Dashboard – Displays multiple reports in one centralized interface.

14. Reports and Analytics

14.1 Employees by Department

Property	Value
Report Name	Employees by Department
Source Type	Table
Table	Employee Test
Type	Pie Chart
Group By	Department
Aggregation	Count

The report displays the number of employees belonging to each department.

14.2 Employees by Location

Property	Value
Report Name	Employees by Location
Table	Employee Test
Type	Bar Chart
Group By	Location
Aggregation	Count

The report displays employee distribution based on location.

14.3 Employee List Report

Property	Value
Report Name	Employee List Report
Table	Employee Test
Type	List
Columns	Employee ID, Employee Name, Email, Department, Location

This report displays complete employee information in list format.

15. Dashboard

15.1 Create Dashboard

Navigation: All → Dashboards → Self Service → Dashboards

Property	Value
Dashboard Name	Employee Analytics Dashboard

Save the dashboard.

15.2 Add Reports to Dashboard

24. Open the required report.
25. Select Share → Add to Dashboards.
26. Select Employee Analytics Dashboard.
27. Click Add.
28. Repeat for Employees by Location and Employee List Report.

The three reports are displayed together on the Employee Analytics Dashboard. Dashboard sharing options can be used according to appropriate access permissions.

16. Testing Strategy

Testing is performed at each major stage of the project.

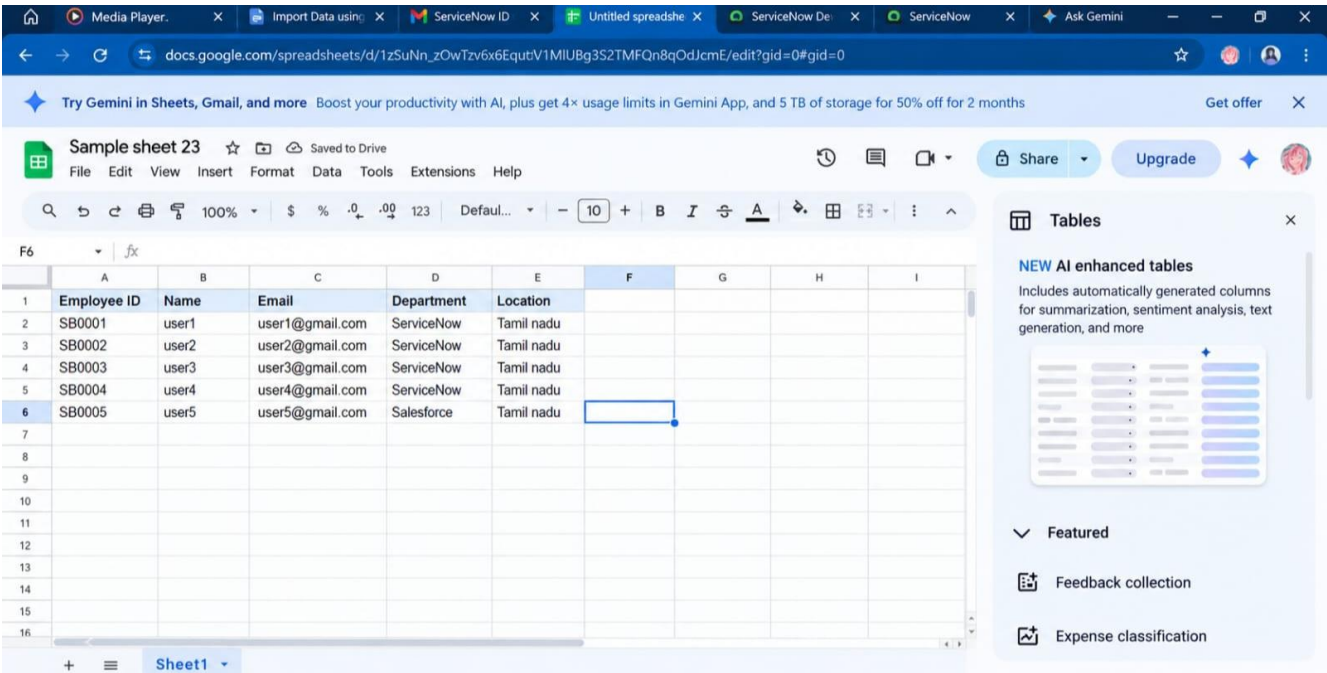
Test Case	Input	Expected Result
Spreadsheet Upload	Valid Excel file	Data loads successfully
Import Set	Employee records	Records appear in Import Set
Field Mapping	Source and target fields	Correct mapping
Initial Transform	New records	Records inserted
Existing Employee	Existing Employee ID	Record updated
New Employee	New Employee ID	Record inserted
Duplicate Import	Same spreadsheet	Duplicate records prevented
Transform History	Completed transformation	Insert/update/ignore status displayed
Department Report	Employee Test data	Department-wise count displayed
Location Report	Employee Test data	Location-wise count displayed
Employee List	Employee Test data	Employee records displayed
Dashboard	Three reports	Reports displayed together

Validation

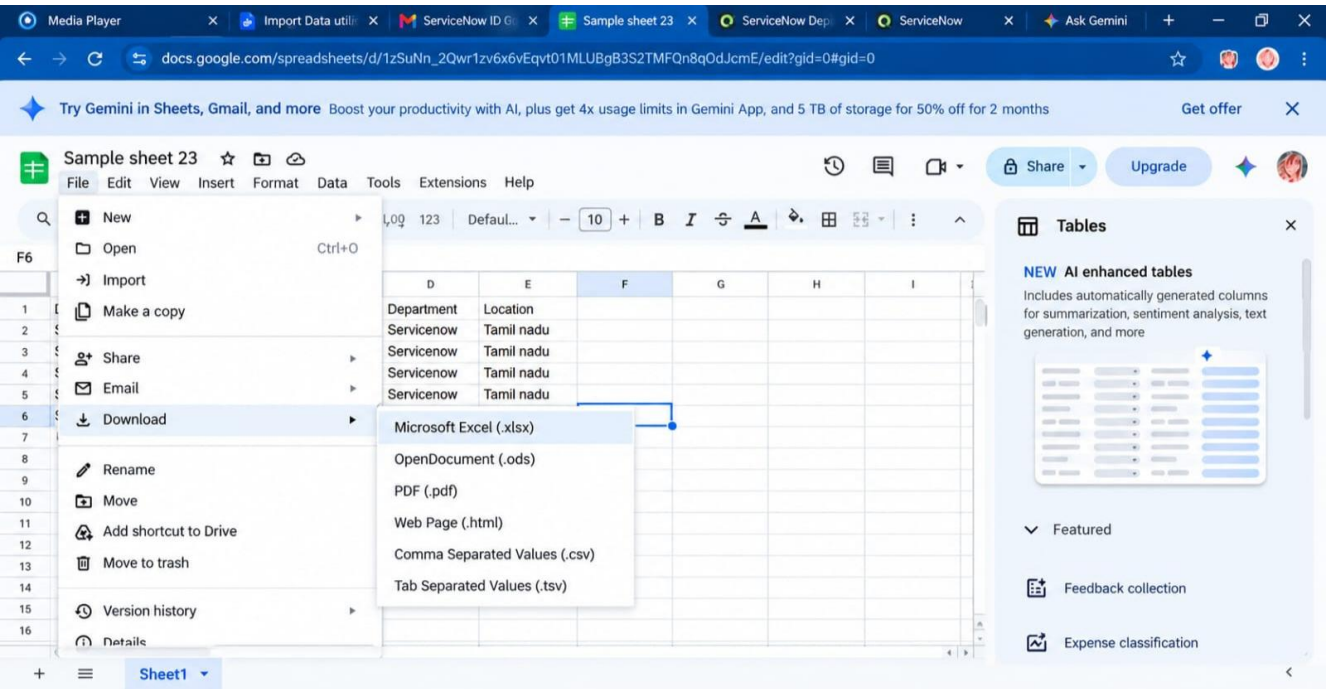
- Imported record count.
- Field values.
- Employee IDs.
- Updated records.
- Newly inserted records.
- Ignored records.
- Transform status.
- Report results.
- Dashboard display.

17. Screenshots / Demo

29. Sample Spreadsheet



30. Employee Test table creation



31. Employee Test field

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D8b99ef8283278710cf259430cead37c%26sysparm_view%3D%26syspar...

servicenow All Favorites History Workspaces Table - Employee Test Search Delete Update Delete All Records

Table Employee Test

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Employee Test Application Global
* Name u_employee_test Remote Table

Columns Controls Application Access

Table Columns for text Search 1 to 6 of 6 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	-40	-40	false
Updated by	String	(empty)	-40	-40	false
Updated	Integer	(empty)	-40	-40	false
Sys ID	Sys ID (GUID)	(empty)	32		false

29. Load Data page

dev410933.service-now.com/now/nav/ui/classic/params/target/u_employee_test.do%3Fsys_id%3D-1

servicenow All Favorites History Admin Employee Test - Create Created Search Submit

Load data

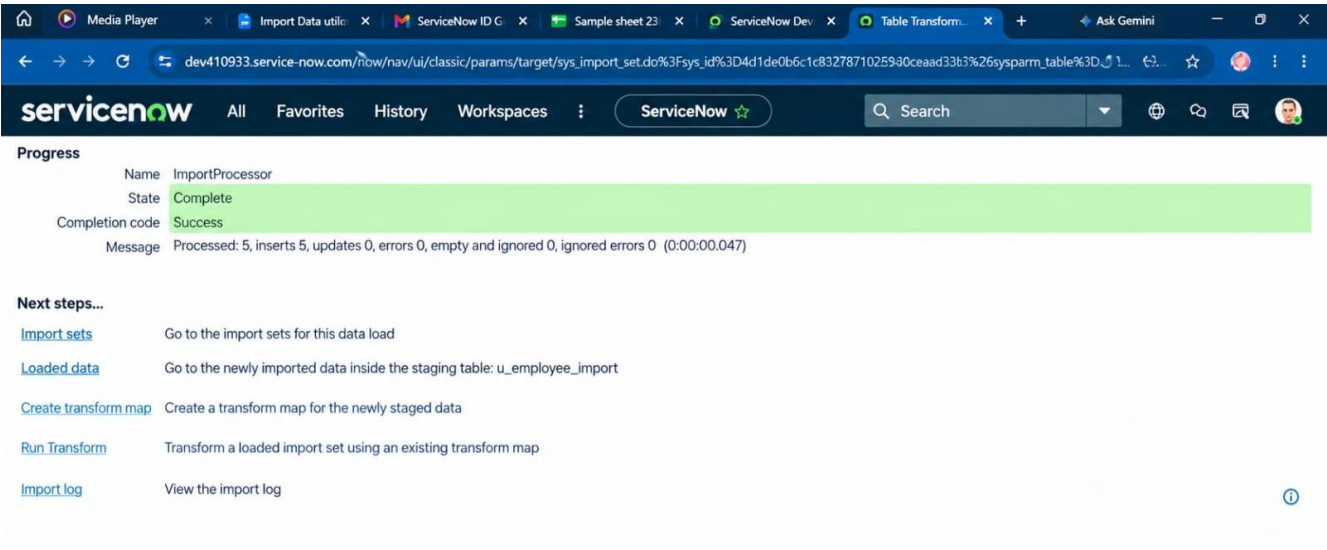
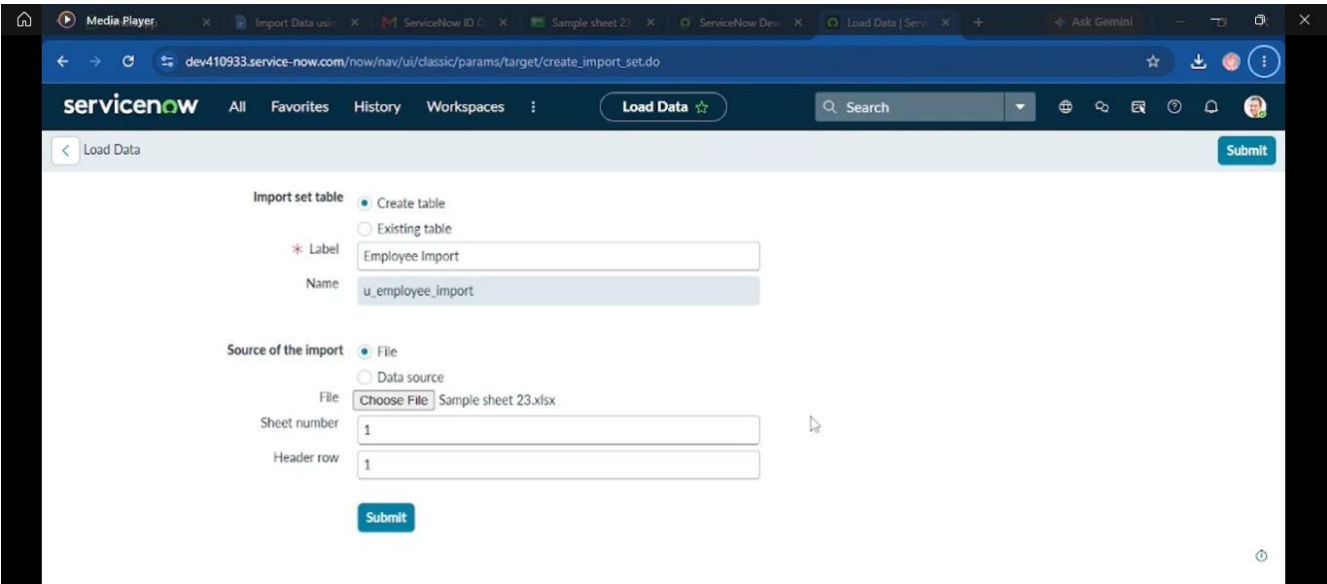
FAVORITES
No Results

ALL RESULTS
System Import Sets
Load Data

Load data from a file for import to begin an import to ServiceNow

Submit

Employee Import table



30. Transform Map configuration

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D-1%26sysparm_query%3Dsource_table%3Du_employee_import

service-now All Favorites History Table Transform Map - New Record Search

Table Transform Map
New record

* Name: Sample Spreadsheet Import

* Source table: Employee Import [u_employee_im...]

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created:

* Target table: Employee Test [u_employee_test]

Order: 100

Run script: ☐

Submit

Related Links

dev410933.service-now.com/now/nav/ui/classic/params/target/mapping_assist.do%3Fsysparm_stack%3Dnc%26sysparm_source_table%3Du_employee_import%26sysp...

service-now All Favorites History Workspaces ServiceNow Search

Mapping Assist Save Cancel

Source: Employee Import

Field Map

Source	Target
Department	Department
Email	Email
Employee ID	Employee ID
Location	Location

Target: Employee Test

Employee Name

Data Viewer

Show: ☒ All Fields ☐ Mapped Fields

Employee Import Viewing: 1 Employee Test Viewing: 1

Field	Value	Field	Value

31. Field Mapping

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3Dfb6fe9832787...

servicenow All Favorites History Table Transform Map - Sample Sprea... Search

Table Transform Map
Sample Spreadsheet Import

coalesce fields

Copy Update Delete

Related Links

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)

Field Maps (4) Transform Scripts

Source field	Target field	Coalesce
☐ u_department	u_department	false
☐ u_email	u_email	false
☐ u_employee_id	u_employee_id	false
☐ u_location	u_location	false

1 to 4 of 4

32. Successful Transform

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3Db26e6e832787102f259430caad3d5

servicenow All Favorites History Table Transform Map - Sample Sprea... Search

Table Transform Map
Sample Spreadsheet Import

Run business rules ☒ Order 100

Enforce mandatory fields No

Copy empty fields ☐

Create new record on empty coalesce fields ☐

Run script ☐

Copy Update Delete

Related Links

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)

Field Maps (5) Transform Scripts

Source field	Target field	Coalesce
--------------	--------------	----------

The screenshot shows the ServiceNow web interface. The browser tabs include Media Player, Import Data util, ServiceNow ID G, Sample sheet 23, ServiceNow Dev, and Table Transform. The URL is dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D1b626e6e83278700f2f59430ceaad3d55%26sysparm_query=... The ServiceNow logo and navigation menu are visible. The main content area shows a transformation job summary:

- Name: Transforming: ISET0010003
- State: Complete
- Completion code: Success
- Message: Transformation complete

Below the summary, there are "Next steps..." with links:

- [ISET0010003](#): Go to the import sets for this data load
- [Transform history](#): Show the transform history, related errors and log
- [Import log](#): View the import log

33. Employee Test records

The screenshot shows the ServiceNow web interface for the "Employee Test" list. The browser tabs include Media Player, Import Data util, ServiceNow ID G, Sample sheet 23, ServiceNow Dev, and Employee Test. The URL is dev410933.service-now.com/now/nav/ui/classic/params/target/u_employee_test_list.do%3Fsysparm_userpref_module%3D7dd9e7c2832787102f259430ceaad355%26sysparm_query=... The interface includes a search bar, a filter dropdown set to "Department", and a "Personalize List" button. The table below displays the data:

Department	Email	Employee ID	Employee Name	Location
Servicenow	user4@gmail.com	SB0004	user4	Tamil nadu
Salesforce	user5@gmail.com	SB0005	user5	Tamil nadu
Servicenow	user2@gmail.com	SB0002	user2	Tamil nadu
Servicenow	user1@gmail.com	SB0001	user1	Tamil nadu
Servicenow	user3@gmail.com	SB0003	user3	Tamil nadu

34. Coalesce configuration

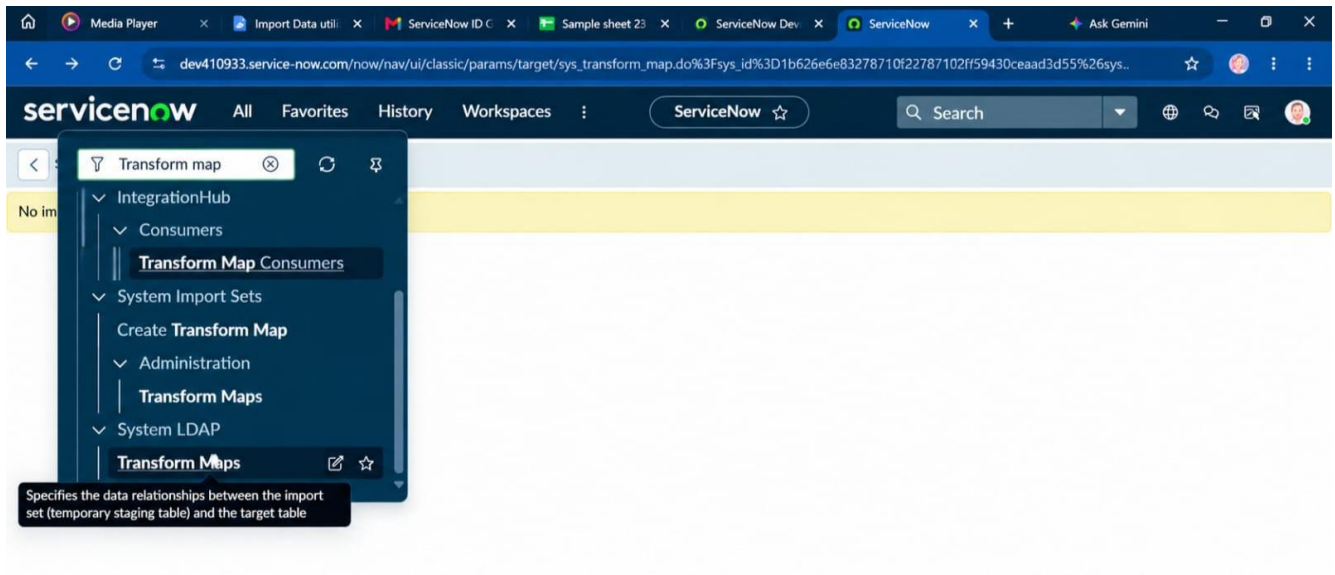


Table Transform Maps							
Name	Source table	Target table	Run business rules	Order	Active	Updated	
Sample Spreadsheet Import	Employee Import [u_employee_import]	Employee Test [u_employee_test]	true	100	true	2026-09-27 04:33:40	

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D-1%26sysparm_query%3Dsource_table%3Du_employee_import

service-now All Favorites History Table Transform Map - New Record Search

Table Transform Map
New record

* Name: Sample Spreadsheet Import

* Source table: Employee Import [u_employee_im...]

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created:

* Target table: Employee Test [u_employee_test]

Order: 100

Run script: ☐

Submit

Related Links

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D1b626e6e83278710f2f59430ceaad3d55%26sysparm_record_target%3Dsys...

Table Transform Map
Sample Spreadsheet Import

Mapping Fields

Validate Fields

Transform Fields

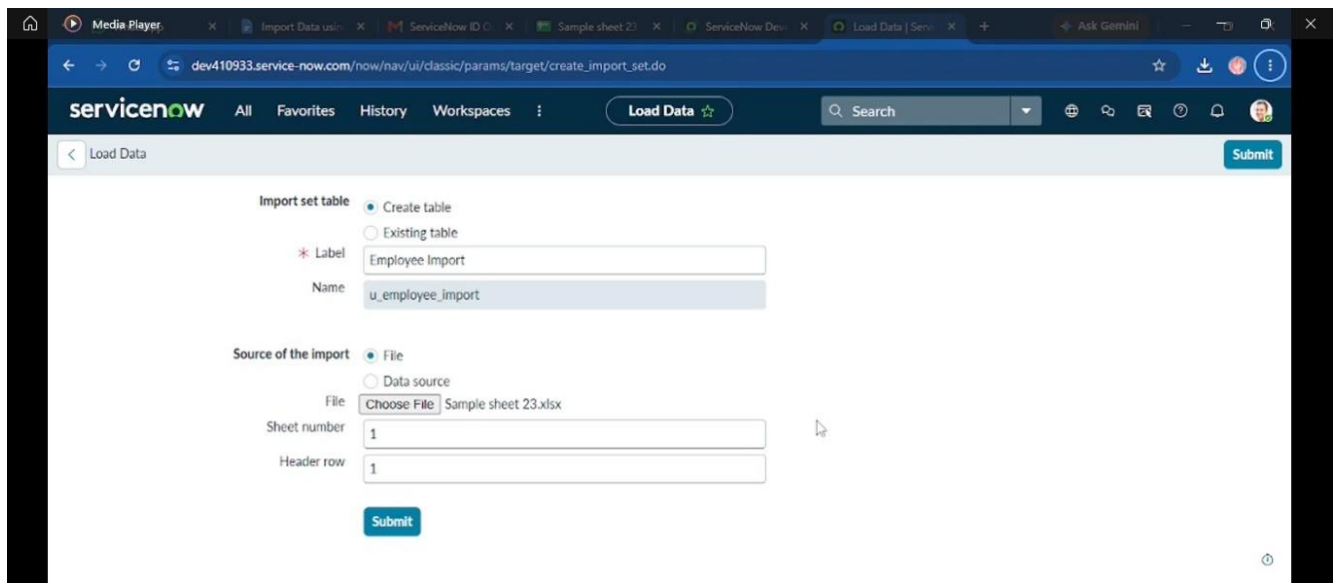
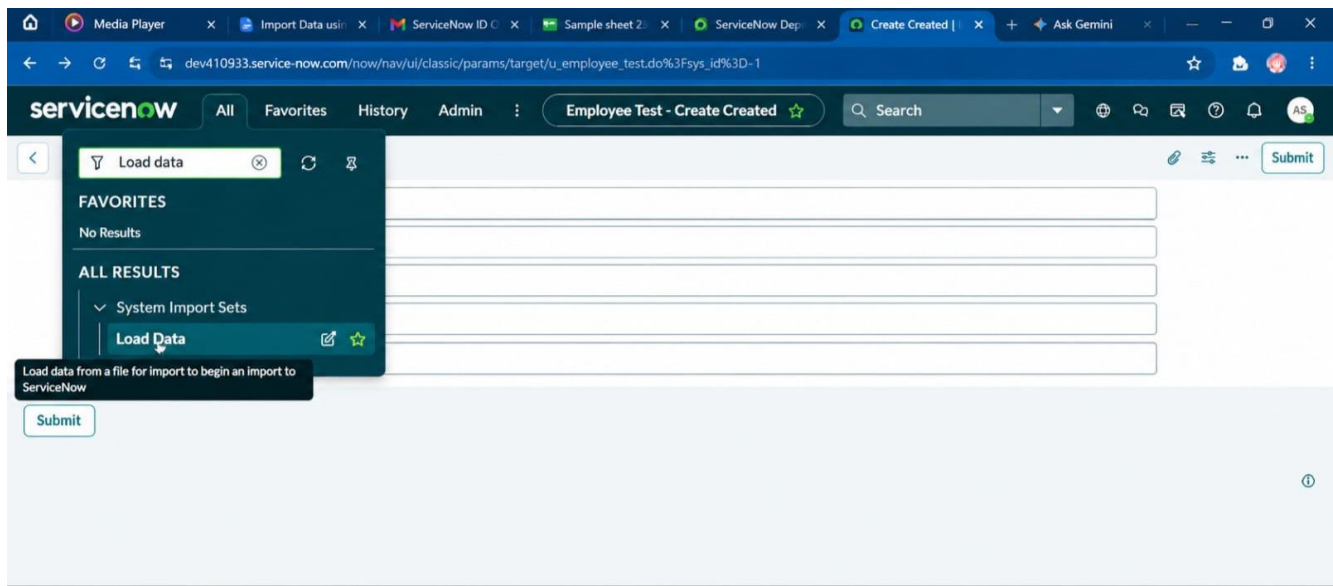
Transform Scripts

Actions on selected rows... New

Source field	Target field	Coalesce
Name	u_employee_name	true
Department	u_department	false
Email	u_email	false
Employee_id	u_employee_id	false
Location	u_location	false

1 to 5 of 5

35. New Excel data



Sample sheet 23

File Edit View Insert Format Data Tools Extensions Help

100% 123 Default... - 10 + B I A

E7 fx Tamil nadu

	A	B	C	D	E	F	G	H	I
1	Employee ID	Name	Email	Department	Location				
2	SB0001	user1	user1@gmail.com	ServiceNow	Tamil nadu				
3	SB0002	user2	user2@gmail.com	ServiceNow	Tamil nadu				
4	SB0003	user3	user3@gmail.com	ServiceNow	Tamil nadu				
5	SB0004	user4	user4@gmail.com	ServiceNow	Tamil nadu				
6	SB0005	user5	user5@gmail.com	Salesforce	Tamil nadu				
7	SB0006	user6	user6@gmail.com	Salesforce	Tamil nadu				
8									
9									
10									
11									
12									
13									
15									
16									
17									

Tables

NEW AI enhanced tables

Includes automatically generated columns for summarization, sentiment analysis, text generation, and more.

Featured

- Feedback collection
- Expense classification

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_import_set.do%3Fsys_id%3D4d1de0b6c1c8327871025930cead33b3%26sysparm_table%3D%3D...

ServiceNow

Progress

Name ImportProcessor

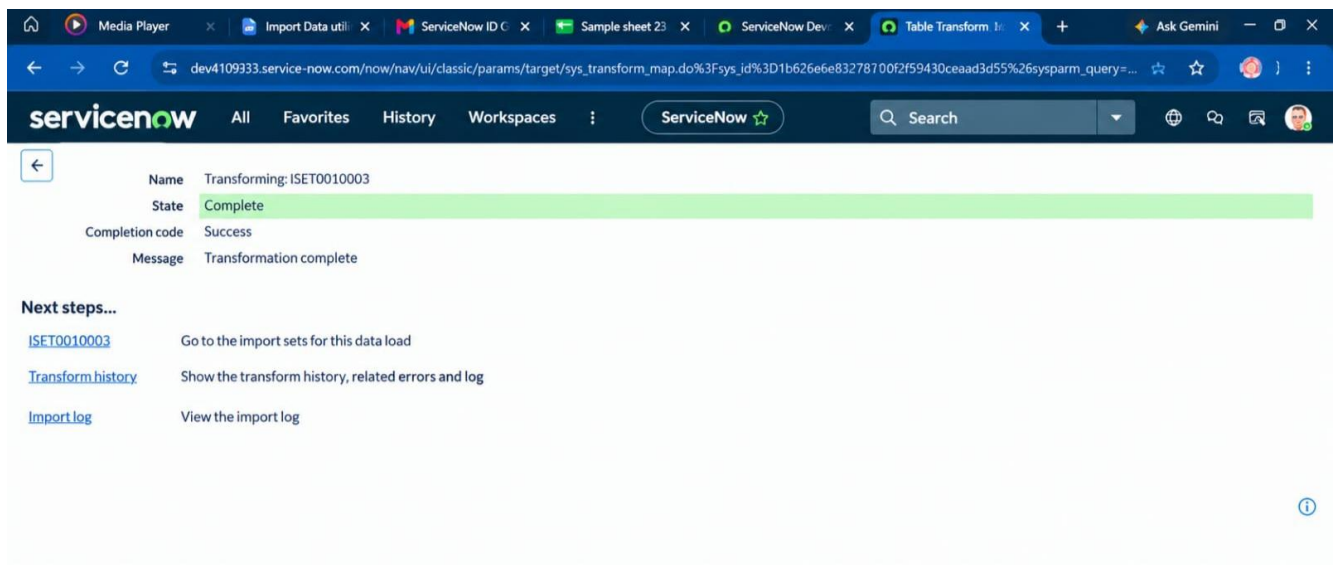
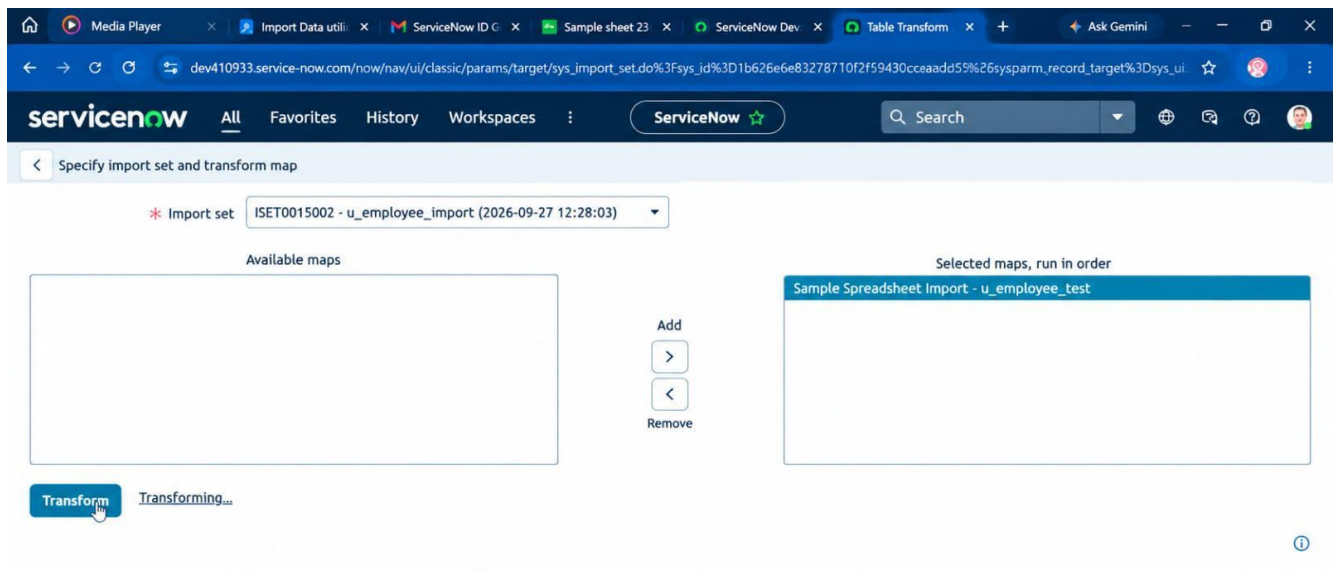
State Complete

Completion code Success

Message Processed: 5, inserts 5, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:00.047)

Next steps...

- [Import sets](#) Go to the import sets for this data load
- [Loaded data](#) Go to the newly imported data inside the staging table: u_employee_import
- [Create transform map](#) Create a transform map for the newly staged data
- [Run Transform](#) Transform a loaded import set using an existing transform map
- [Import log](#) View the import log

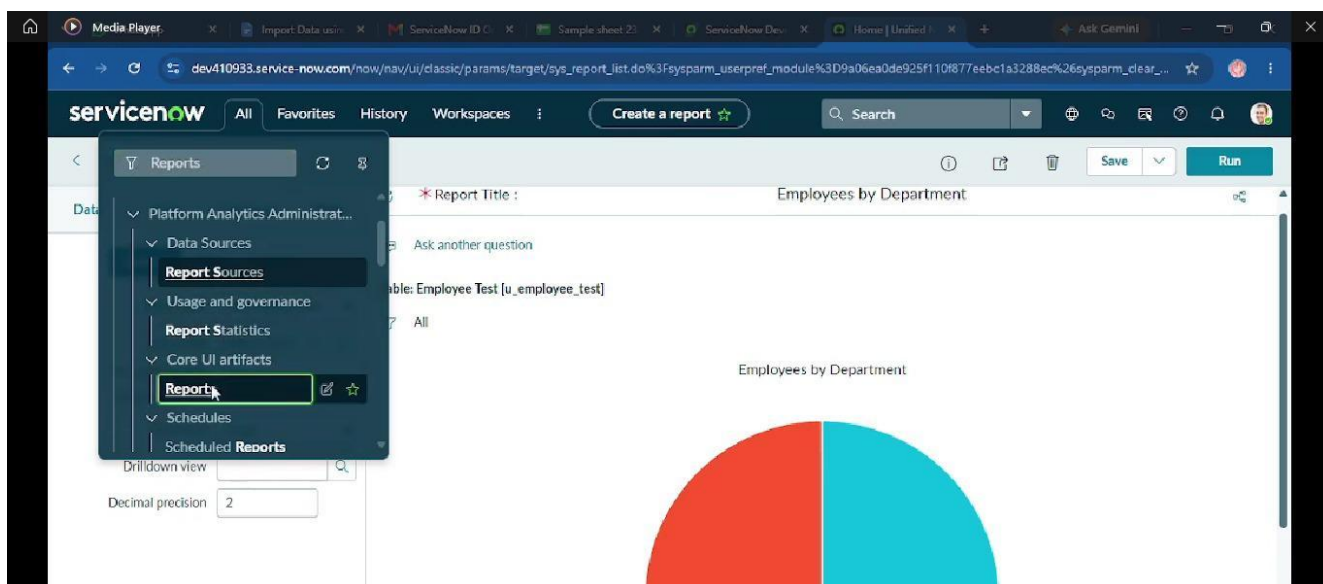


Transform History

The screenshot shows the 'Transform Histories' page in ServiceNow. The table lists the following record:

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors
2026-09-27 06:51:42	Complete	2026-09-27 06:51:42	0 Seconds	ISSET00100003	Employee Import (u_employee_import)	6	1	0	5	0	Sample Import

36. Employees by Department report



dev410933.service-now.com/now/nav/ui/classic/params/target/sys_report_list.do%3Fsysparm_userpref_module%3D9a06ea0de925f110f877eebc1a3288ec%26sysparm_clear...

servicenow All Favorites History Workspaces Reports Search

for text Search Actions on selected rows... New

Title	Table	Type	Field Name	Created by	Updated
KPI - Average Work Effort for Resolving...	Incident Time Worked [incident_time_worked]	Pivot Table	inc_category	glide.maint	2011-11-30 14:35:43
30/60/90 Day Desired State Task Aging	Follow On Task [cert_follow_on_task]	Horizontal bar	aging_level	admin	2013-04-18 13:57:43
30/60/90 Day Task Aging	Follow On Task [cert_follow_on_task]	Horizontal bar	aging_level	admin	2013-04-18 13:30:43
abc	label [test_2]	Single score		admin	2020-03-06 18:38:27
Achieved SLAs by Type	Task SLA [task_sla]	Vertical bar	sla.type	glide.maint	2012-11-29 05:42:30
Active Change Requests	Change Request [change_request]	List	active	glide.maint	2005-06-21 18:17:56
Active Changes > 7 days	Change Request [change_request]	Single score		admin	2020-11-11 11:43:52

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_report_template.do%3Fsysparm_create%3Dtrue

servicenow All Favorites History Workspaces Create a report Search

Create a report Save Run

Data > Type > Configure > Style

* Report Title : Employees by Department

* Report name: Employees by Department

* Source type: Table

* Table: Employee Test [u_employee_test]

Description: There is no description for this table. To add a description, please contact your admin.

Table: Employee Test [u_employee_test]

All

Create your report with Analytics Q&A

Ask for information. You can give simple filtering conditions. You get the answer with an appropriate visualization.

What do you want to see? Ask

How can I improve my results?

Media Player, Import Data usin, ServiceNow ID C, Sample sheet 23, ServiceNow Des, Create a report, Ask Gemini

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_report_template.do%3Fsysparm_create%3Dtrue

servicenow All Favorites History Workspaces Create a report Search

Create a report Save Run

Data > Type > Configure > Style

pie chart

2 Report types found by search
Select one of the valid types

Pies and Donuts
Pies and Donuts show the proportions that make up a whole.

Report Title : Employees by Department

Type a question about your data
What do you want to see? Ask How can I improve my results?

To modify the current report, use the left panel or Edit Condition.

Table: Employee Test [u_employee_test]

All

Employees by Department

Employee ID	Employee Name	Department	Email	Location
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0006	user6	Salesforce	user6@gmail.com	Tamil nadu
SB0005	user5	Salesforce	user5@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu

Media Player, Import Data usin, ServiceNow ID C, Sample sheet 23, ServiceNow Des, Create a report, Ask Gemini

dev410933.service-now.com/now/nav/ui/classic/params/target/sys_report_template.do%3Fsysparm_create%3Dtrue

servicenow All Favorites History Workspaces Create a report Search

Create a report Save Run

Data > Type > Configure > Style

Group by
Department
Additional group by
☐ Display data table
Configure function field

Aggregation
Count

Set Value Formatting

Max number of groups
System Default

☒ Show Other

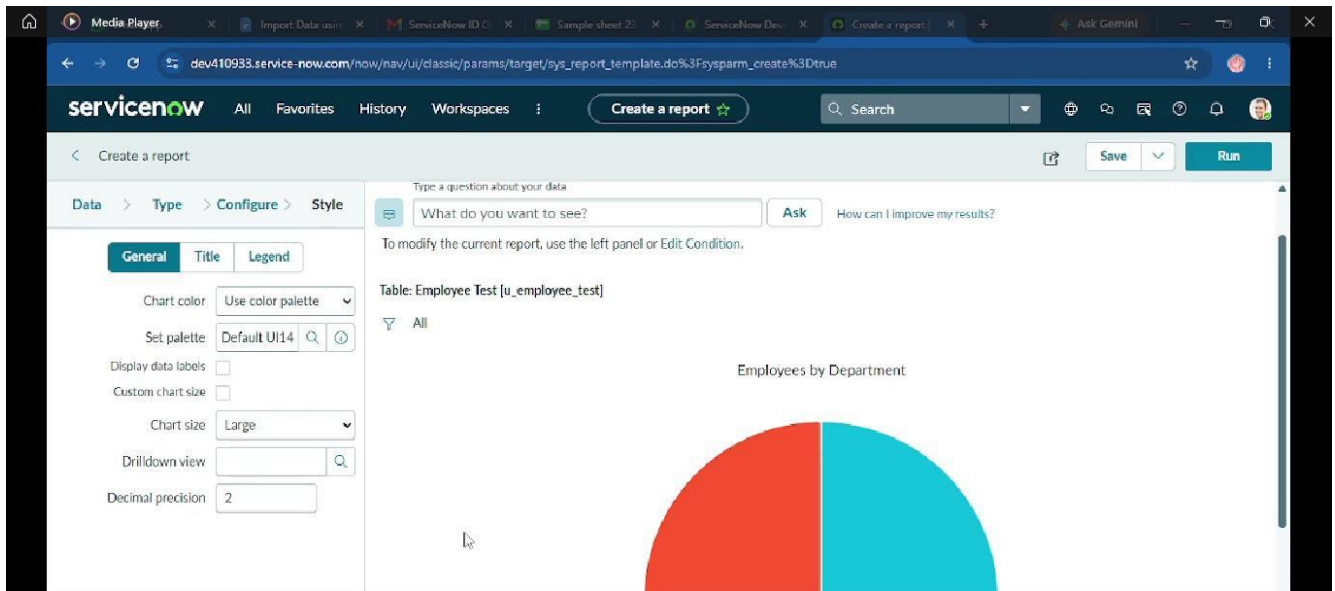
Type a question about your data
What do you want to see? Ask How can I improve my results?

To modify the current report, use the left panel or Edit Condition.

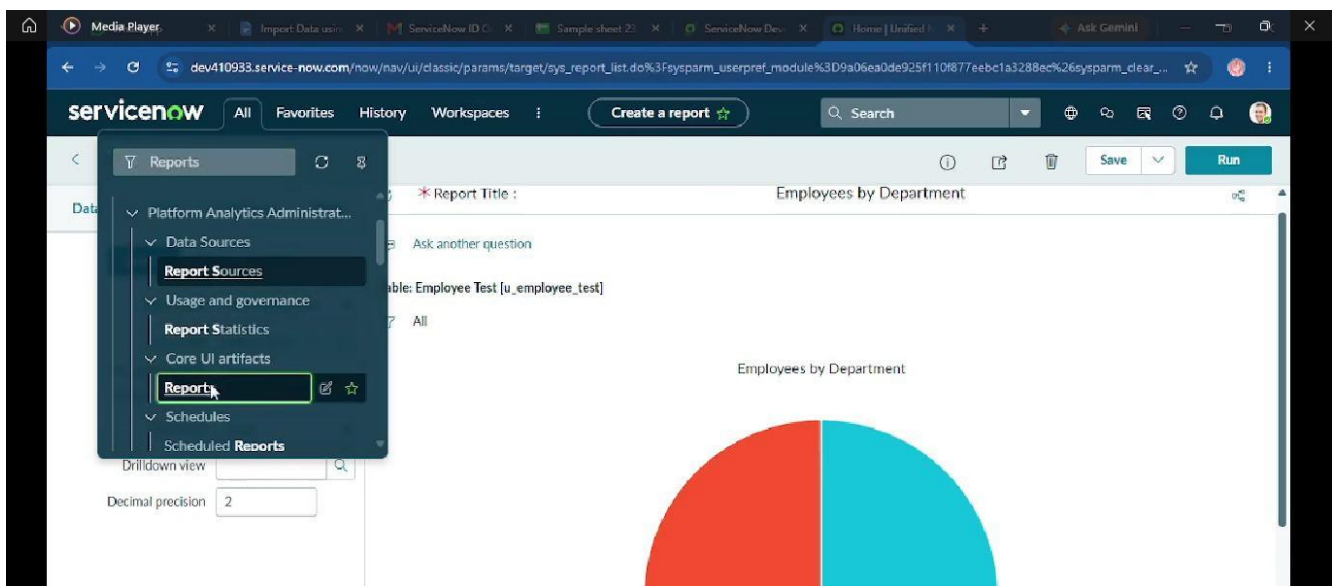
Table: Employee Test [u_employee_test]

All

Employees by Department



37. Employees by Location report



38. Employee List Report

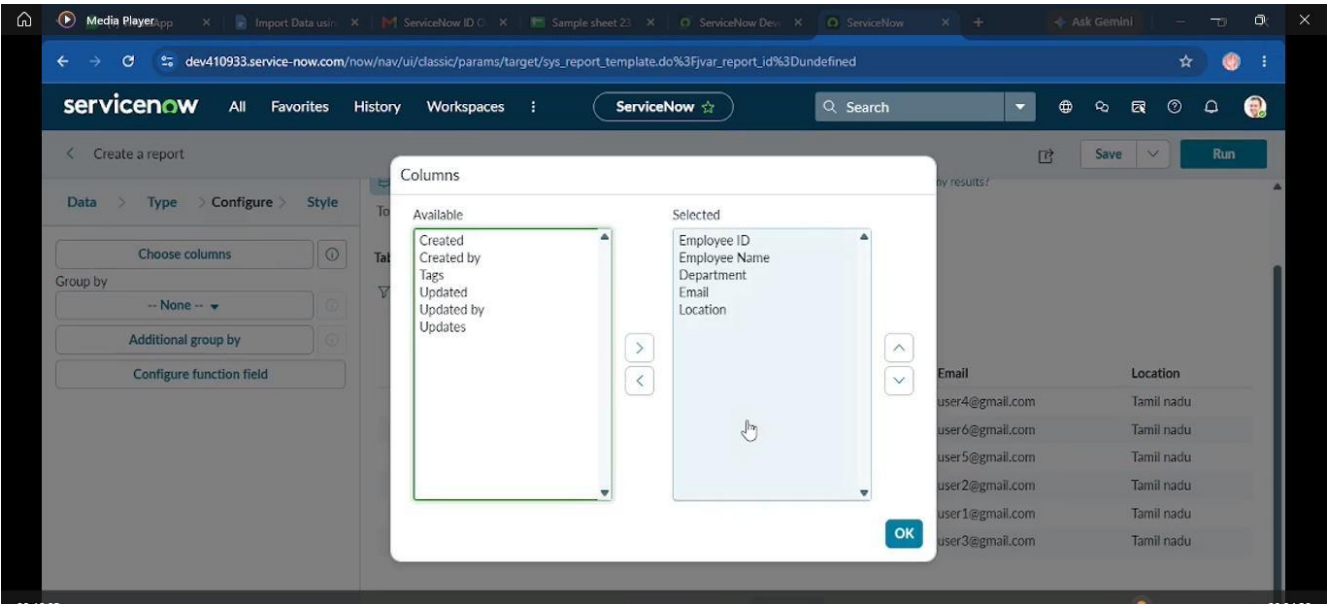
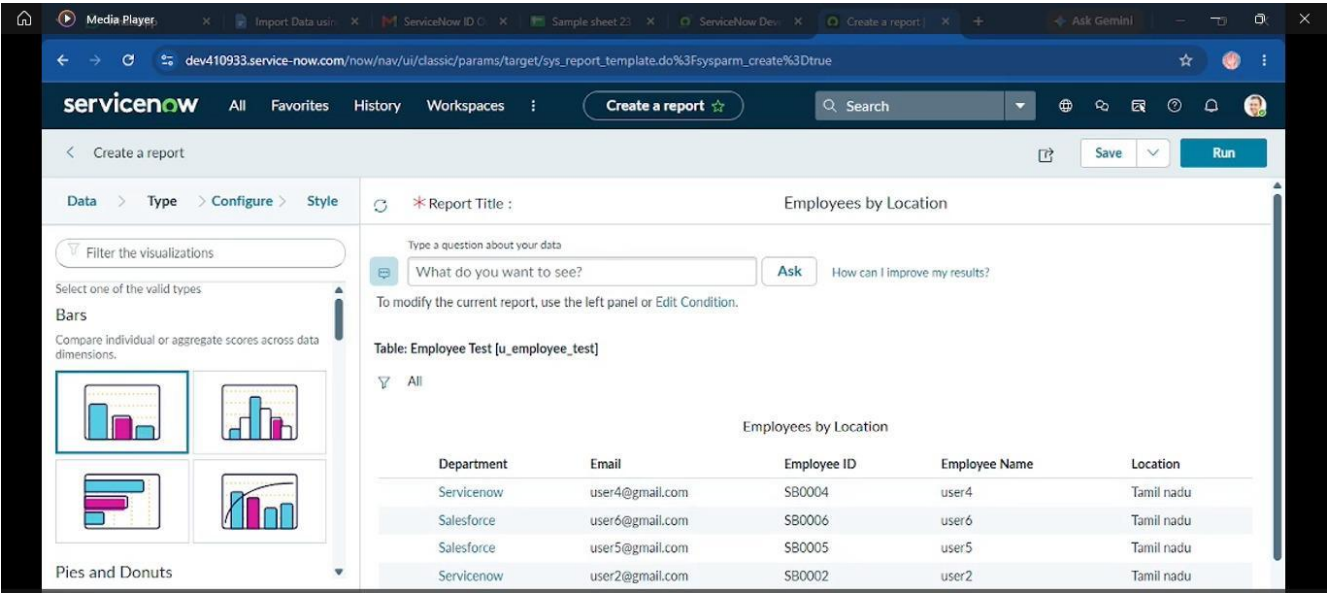


Table: Employee Test [u_employee_test]

Employee List Report

Employee ID	Employee Name	Department	Email	Location
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0006	user6	Salesforce	user6@gmail.com	Tamil nadu
SB0005	user5	Salesforce	user5@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu
SB0001	user1	ServiceNow	user1@gmail.com	Tamil nadu
SB0003	user3	ServiceNow	user3@gmail.com	Tamil nadu

39. Employee Analytics Dashboard

Access Control (ACL)

Build Agent

GO FURTHER

Media Player: dev410933.servicenow.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_query%3DnameSTARTSWITHpa_dashboards%26sysparm_first_row%3D1...

servicenow All Favorites History Workspaces Access Controls Search

Access Controls Name Search Actions on selected rows...

All > Name starts with pa_dashboards

	Name	Decision Type	Operation	Type	Active	Updated by	Updated
	pa_dashboards	Allow If	write	record	true	admin	2019-03-25 09:32:03
	pa_dashboards	Allow If	write	record	true	admin	2016-04-19 03:29:14
	pa_dashboards	Allow If	read	record	true	admin	2019-03-25 09:34:18
	pa_dashboards	Allow If	create	record	true	admin	2026-02-25 07:30:56
	pa_dashboards	Allow If	report_view	record	true	admin	2022-11-16 22:19:19
	pa_dashboards	Allow If	delete	record	true	admin	2016-04-19 03:26:55
	pa_dashboards	Allow If	query_range	record	true	glide.maint	2025-11-05 15:23:15
	pa_dashboards	Allow If	read	record	true	admin	2018-03-01 07:27:17
☐	pa_dashboards	Allow If	create	record	true	admin	2023-12-20 01:56:39
	pa_dashboards	Allow If	delete	record	true	admin	2019-03-25 09:35:29
	pa_dashboards.*	Allow If	query_range	record	true	glide.maint	2025-11-05 15:27:50
	pa_dashboards.group	Allow If	write	record	true	admin	2016-07-28 06:29:53
	pa_dashboards.owner	Allow If	write	record	true	admin	2021-11-04 08:32:01

Media Player: dev410933.servicenow.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D0e17790531331102485ddeef7b127c%26sysparm_record_target%3Dsys_se...

servicenow All Favorites History Admin Access Control - pa_dashboards Search

Access Control pa_dashboards Update Delete

* Type record Application Global ⓘ

* Operation create ⓘ ⓘ

Admin overrides ☒

Active ☒

Advanced ☐

Decision Type Allow If

Protection policy -- None --

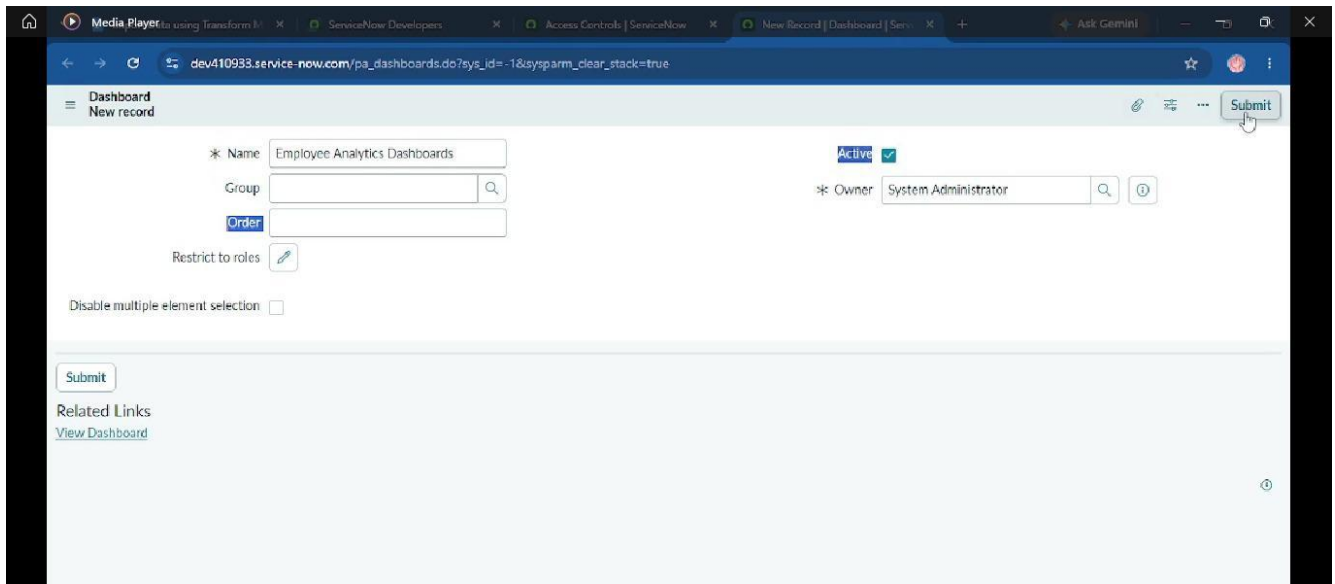
* Name Dashboard [pa_dashboards] -- None --

Description

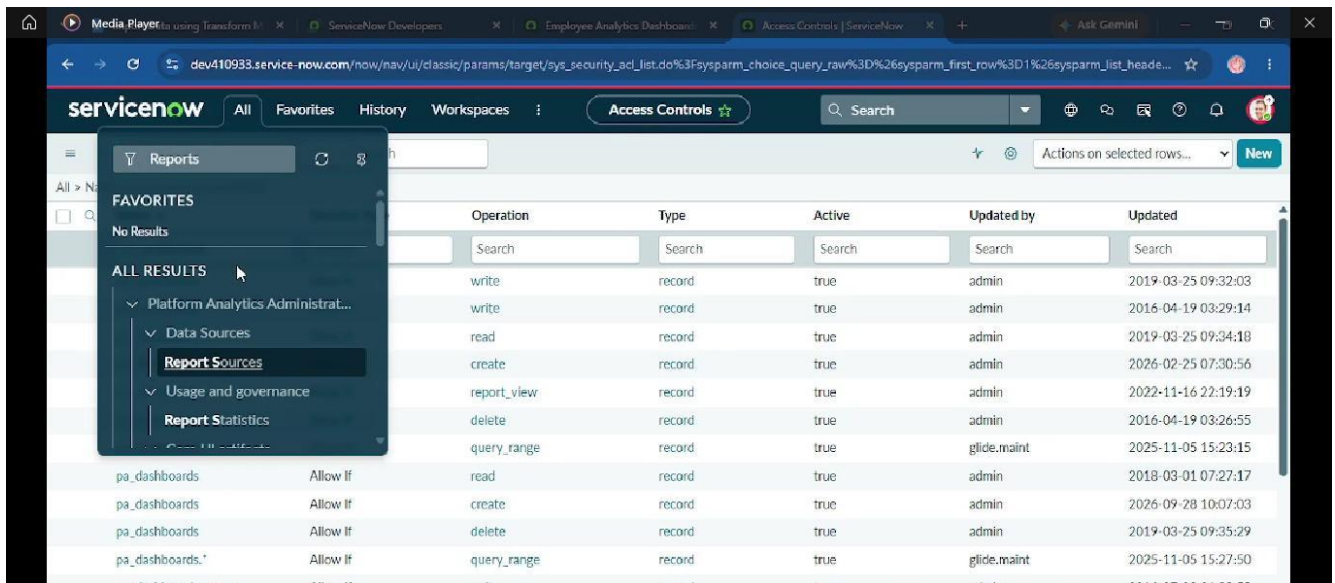
Conditions

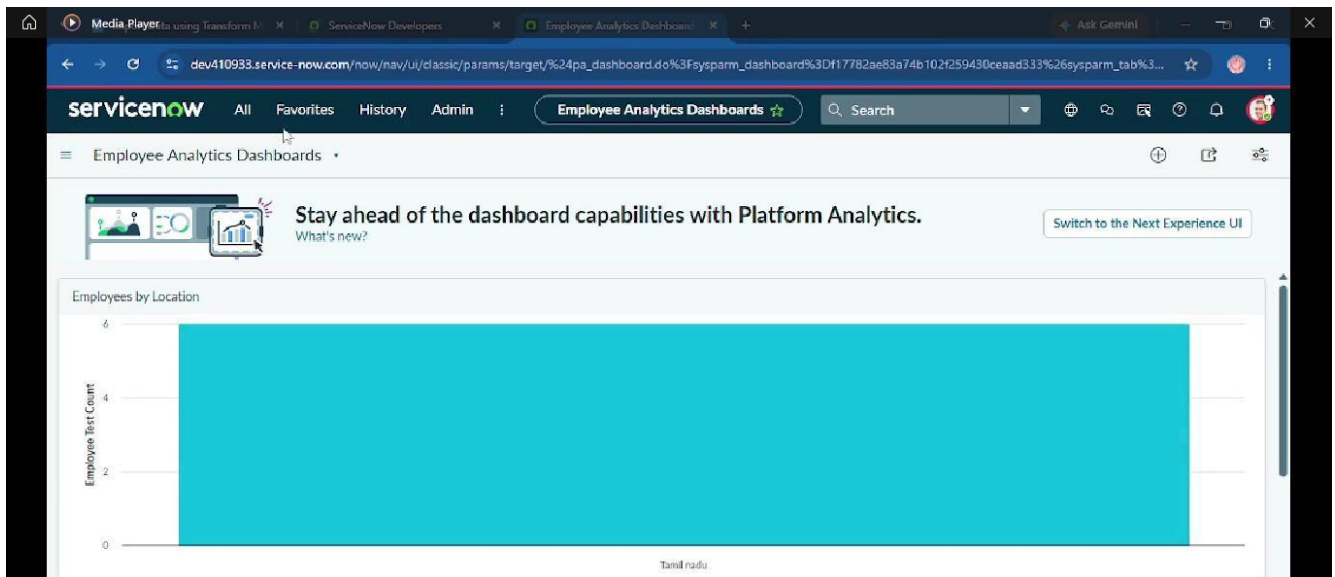
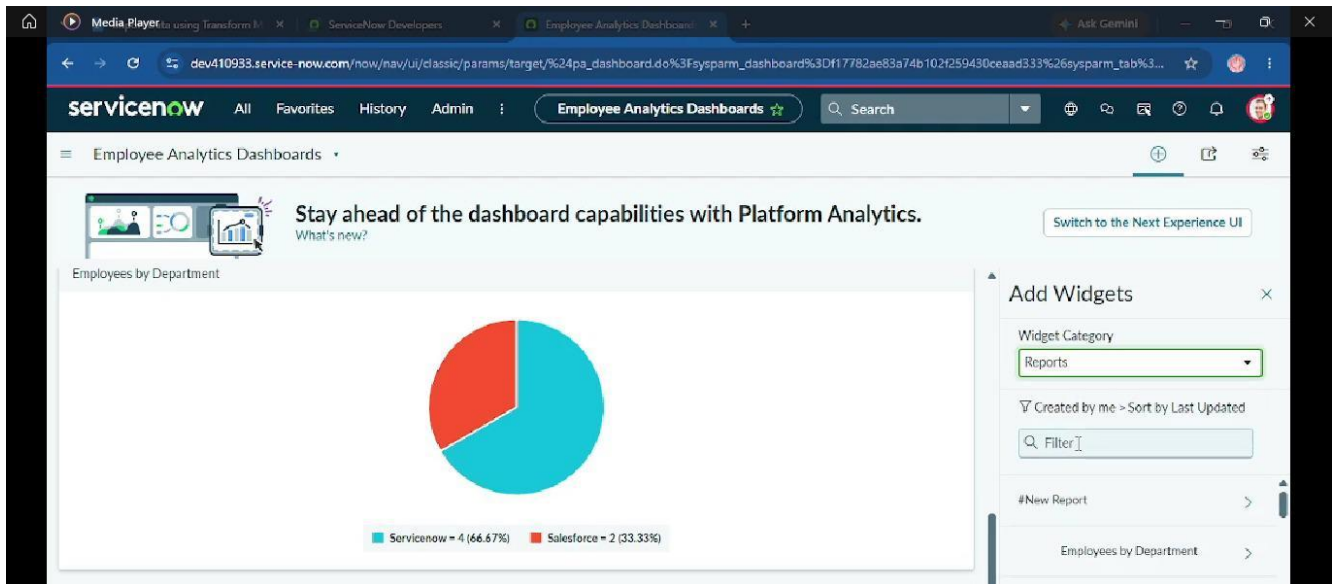
Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.



Add Reports to Dashboards





The screenshot shows the ServiceNow Employee Analytics Dashboard. The main content area displays an 'Employee List Report' table with the following data:

Employee ID	Employee Name	Department	Email	Location
SB0001	user1	ServiceNow	user1@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu
SB0003	user3	ServiceNow	user3@gmail.com	Tamil nadu
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0005	user5	Salesforce	user5@gmail.com	Tamil nadu
SB0006	user6	Salesforce	user6@gmail.com	Tamil nadu

The sidebar on the right, titled 'Add Widgets', shows a 'Widget Category' dropdown set to 'Reports'. Below it, there's a filter section with a search bar and a list of reports including '#New Report' and 'Employees by Department'.

The screenshot shows a dashboard layout with three widgets. The 'Employees by Location' widget is a bar chart showing 'Employee Test Count' for 'Tamil nadu'. The 'Employee List Report' widget is a table showing employee details. The 'Employees by Department' widget is a pie chart showing the distribution of employees by department.

Department	Count	Percentage
ServiceNow	4	66.67%
Salesforce	2	33.33%

The 'Share' sidebar on the right shows options to 'Add groups, users and roles' and a list of roles: 'All', 'Groups', 'Users', and 'Roles'. A message states: 'This dashboard hasn't been shared with anyone'.

Complete Demo Flow

Spreadsheet → Load Data → Import Set → Transform Map → Field Mapping → Transform → Employee Test → Coalesce Testing → Reports → Dashboard

18. Known Issues

40. Incorrect Field Mapping – Incorrect mappings can store data in the wrong target fields.
41. Missing Values – Missing employee information may result in incomplete records or transformation problems.

- 42. Duplicate Employee IDs – Incorrect IDs can affect Coalesce processing.
- 43. Invalid Spreadsheet Data – Incorrect formats or invalid values may cause transformation errors.
- 44. Permission Restrictions – Some ServiceNow functions require appropriate roles and permissions.
- 45. Dashboard Access – Users without required permissions may not be able to view or modify dashboards.
- 46. Large Spreadsheet Files – Large datasets may require additional processing time.

19. Future Enhancements

- Automated Data Import – Schedule regular employee data imports.
- Advanced Data Validation – Validate Employee ID, email, department, location and mandatory fields.
- Transform Scripts – Add advanced transformation logic and business rules.
- Automated Notifications – Notify administrators when imports succeed or fail.
- Enhanced Dashboard – Add total employees, new employees, updated employees, success/error counts and distribution widgets.
- External System Integration – Replace manual spreadsheets with automated HR-system integration.
- Improved Error Handling – Provide better mechanisms for identifying and reporting failed records.

20. Project Outcome

The project demonstrates an end-to-end employee data import and management process using ServiceNow. Employee data is prepared in Excel, loaded into an Import Set staging table, transformed into the Employee Test target table, validated, and analyzed through reports and a dashboard.

- Spreadsheet data is successfully loaded into ServiceNow.
- Import Set data is stored in a staging table.
- Transform Map transfers data to the Employee Test table.
- Source and target fields are mapped correctly.
- New employee records are inserted.
- Coalesce identifies existing employees.
- Existing records are updated instead of duplicated.
- Repeated records can be prevented from unnecessary insertion.
- Transform History validates transformation results.
- Reports provide employee data analysis.
- Dashboard provides a centralized view of reports.

21. Conclusion

This project successfully implements an employee data management solution using ServiceNow Import Sets and Transform Maps. It demonstrates how structured employee information can be transferred from an external Excel spreadsheet into ServiceNow in an efficient and controlled manner. The project also demonstrates field mapping, Coalesce-based duplicate prevention, data validation, reporting and dashboard creation. Overall, the solution provides a practical workflow for managing bulk employee data within the ServiceNow platform.

Demo Link: <https://drive.google.com/file/d/11y44IzATHLhhasM6nbEQ4exVWkC3IIZG/view?usp=drivesdk>